ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
School of Information and Communication Technology

PROJECT PROPOSAL

Course: Web Mining (Khai phá Web) – IT4868
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Skill Demand in Vietnam's IT and Business Analyst Job Postings: A Text Mining Study, 2020–2025

Supervisor: TS. Dương Quang Huy
Team: Group 1

No.	Full name (Họ và tên)	Student ID (MSSV)
1	Nguyễn Trần Minh Trí	202400116
2	Tô Duy An	202416653
3	Kiều Việt Sơn	202416750
4	Nguyễn Ngọc Tuấn Anh	202416659

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Abstract

Employers describe job requirements in free text, using inconsistent spellings, abbreviations, and a mix of Vietnamese and English for the same skill. This makes it difficult to compare skill demand across roles or to monitor how requirements change over time. This project proposes a reproducible Web Mining pipeline that identifies Information Technology (IT) and Business Analyst (BA) postings in a research-permitted Vietnamese job-advertisement corpus, extracts skill mentions, maps aliases to canonical skill names, and analyzes skill prevalence and co-occurrence by role, location, and period (target window 2020–2025, subject to verified data coverage).

Skills will be extracted with an interpretable bilingual dictionary baseline and compared with a PhoBERT-based sequence-labeling model; a manually annotated gold sample will be used to measure precision, recall, and F1. The normalized skills will then be summarized as posting-level prevalence and as separate IT and BA skill co-occurrence networks. Deliverables include a cleaned analysis dataset, a versioned skill vocabulary, extraction and evaluation code, and visualizations. Findings will describe the sampled postings and will not be presented as a complete measure of the Vietnamese labor market.

Keywords: web mining, text mining, information extraction, skill extraction, job postings, co-occurrence network, PhoBERT, Vietnam.

1. Introduction

1.1. Background and motivation

Online job advertisements are a rich source of evidence about how employers express role requirements. A single paragraph may mention programming languages, frameworks, cloud platforms, analysis methods, certifications, and interpersonal capabilities. In Vietnam, postings for technology roles are frequently written in Vietnamese, in English, or in a mixture of both, and the same skill may appear under several surface forms (for example *JS*, *Javascript*, and *JavaScript*; and *business analysis*).

A simple word count therefore overstates some skills, misses aliases, and makes comparisons between job families unreliable. Students, training programs, and job seekers benefit from a transparent and evaluated description of which skills are requested together and how this differs between IT and BA roles.

1.2. Problem statement

Job requirements are mostly unstructured and vary in wording and language across employers and platforms. Without a consistent method for identifying and normalizing skills, comparisons of demand by role, location, or time can be misleading. This project assesses

whether a practical extraction and normalization pipeline can produce sufficiently accurate skill data for descriptive analysis of Vietnamese IT and BA postings, and uses that data to show how skill combinations differ across job families.

2. Aim, Objectives, and Research Questions

2.1. Aim

To develop and evaluate a reproducible pipeline for extracting, normalizing, and analyzing skills in Vietnamese IT and BA job postings.

2.2. Objectives

1. Select a research-permitted Vietnamese job-posting dataset and define a focused sampling frame with verified date coverage.
2. Identify IT and BA postings using transparent title rules, validated on a manually labeled sample, with an optional machine-learning classifier.
3. Extract skill mentions from posting text using a bilingual skill vocabulary and a neural sequence-labeling model.
4. Normalize aliases and spelling variants to canonical skill labels while keeping the original mentions for auditability.
5. Evaluate extraction quality against a manually annotated gold sample using precision, recall, and F1.
6. Compare skill prevalence and co-occurrence across job families, broad locations, and time periods where the data supports it.

2.3. Research questions

RQ1. Which skills are most frequently requested in Vietnamese IT and BA postings during the study period?

RQ2. Which skills tend to appear together, and how do these combinations differ between IT and BA roles?

RQ3. Where location and date coverage are adequate, how does observed skill demand vary across broad locations and years?

RQ4. How accurately do the dictionary baseline and the neural model identify skill mentions in a manually reviewed sample?

These are descriptive questions; the project makes no causal claims and does not forecast future employment outcomes.

3. Scope and Delimitations

- **Geography:** jobs advertised for work in Vietnam; national results first, then broad locations (Hà Nội, TP. Hồ Chí Minh, Đà Nẵng, other/unspecified) when sample sizes allow.
- **Time:** target January 2020 – December 2025. The final window will follow the chosen source's verified coverage; if historical years are missing, the study is shortened or reframed as a snapshot.
- **Job families:** IT (software, data/AI, QA/testing, infrastructure/cloud/security) and BA (business analyst, IT/business systems analyst, process analyst). Boundary roles such as Data Analyst, Product Owner, and Project Manager will be explicitly included or excluded in a documented role-mapping table.
- **Language:** Vietnamese, English, and mixed postings; the language of each posting is recorded and analyzed.
- **Outcome:** percentage of postings requesting each skill and co-occurring skill combinations. Salary and experience are optional and used only if consistently available.
- **Out of scope:** inferring a person's qualifications, predicting hiring outcomes, ranking candidates, or treating advertisement counts as total employment demand.

4. Data Plan

This is just the draft we made, after trying this approch, we may alter it.

4.1. Candidate data sources

Table 1 lists the candidate sources. A dataset is not automatically reusable just because it is publicly downloadable; license, provenance, and coverage will be verified before selection. Existing datasets are preferred for historical coverage; manual collection from job boards (Section 4.2) is the supplementary source and the fallback for a recent snapshot.

Table 1: Candidate data sources

Source	Potential use	Checks / limitations
VietJobs dataset [1]	Primary candidate; reported 48,092 Vietnamese job ads with title, category, salary, and skill fields	Confirm release location, license, date range, duplicate policy, and availability of full descriptions
VietJobs-37K / VieJobBERT [2]	Role-classification benchmark and supplementary Vietnamese text (37,274 postings, 60 occupation labels)	Not designed as a longitudinal skills dataset; check Zenodo license and fields; silver labels are keyword-derived
Techmap / JobDataFeeds [3]	Historical postings since January 2020 with title, description, location, skills	Paid product; confirm academic license, redistribution limits, and exact coverage
University / instructor archive	Preferred if available	Obtain written permission, collection dates, and field definitions
TopCV (manual collection)	Primary board for manual collection; large Vietnamese board with IT and BA categories, full job descriptions, location, and posting/expiry dates	Verify Terms of Service and robots.txt; active postings only (no history); ranking and platform bias
ITviec / TopDev (manual collection)	IT-focused boards; skill tags and full descriptions, many bilingual postings	Verify terms; IT-heavy, few BA postings; some content may need login (not used if so)
VietnamWorks / CareerViet (manual collection)	Optional extra boards to balance BA postings and locations	Verify terms; check that fields match other boards before merging
LinkedIn (optional, manual only)	Small validation sample of English-language postings	User agreement restricts automated collection and most content is login-gated; no scripts, no login bypass; dropped if terms do not allow research use

Selection sequence. (1) Ask the supervisor or library for an existing permitted archive; (2) inspect the VietJobs and VietJobs-37K releases and terms; (3) request a sample and academic terms from the commercial provider if budget allows; (4) in parallel, run the manual collection of Section 4.2 for a recent snapshot from boards whose terms allow it; (5) otherwise narrow the period to a source that can be accessed lawfully. Sources will

not be merged until their fields, dates, and collection methods are comparable.

4.2. Manual collection from job boards

Public datasets may not cover 2020–2025 with full descriptions and dates. To ensure that the project always has usable text, the team will build a small, documented snapshot corpus by manual or semi-manual collection from Vietnamese job boards. Because boards show mainly active postings, this corpus supports prevalence, co-occurrence, and location analyses for the collection period only. Year-by-year trends use it only where the board shows a reliable posting date, and otherwise rely on the historical dataset.

Source priority. TopCV and ITviec/TopDev first (large Vietnamese boards with full descriptions); VietnamWorks/CareerViet to add BA postings; LinkedIn only as an optional, manually collected validation sample.

Compliance protocol.

1. Before collecting from a board, read its Terms of Service and `robots.txt`, and record the date, the URL of the terms, and the decision (allowed / not allowed / unclear) in the source register. Unclear is treated as not allowed unless the supervisor or the site gives written permission.
2. Use only public pages. No login, no CAPTCHA or rate-limit circumvention, no private or applicant-side data.
3. Collection is done by a team member browsing the listing pages and saving each page or copying fields into a structured sheet. Where the terms permit scripts, a single-threaded script with a descriptive user agent and delays of several seconds between requests may be used. Collection stops if a site blocks or warns the client.
4. Raw pages stay private, are not redistributed, and are deleted or archived according to the course and supervisor rules; only aggregated results are published.

Sampling frame. Searches use a fixed query list in Vietnamese and English (e.g. *Business Analyst*, *phân tích nghiệp vụ*, *Software Engineer*, *Data Engineer*, *QA/Tester*, *DevOps*, *Cloud/Security*) with location filters for Hà Nội, TP. Hồ Chí Minh, Đà Nẵng, and other. Collection is quota-based (a target number of postings per role family × city × board, with BA oversampled relative to its natural share) so that groups are large enough to compare. Query terms, filters, collection dates, and counts are logged for every run so that the sample can be described and repeated.

Collected fields.

Table 2: Fields captured during manual collection

Field	Purpose	Handling
Source board, URL, collection date	Provenance and repost checks	URL stored as a hash in the shared dataset
Job title	IT/BA family	Kept verbatim
Description sections	Skill extraction	Requirements, responsibilities, and benefits saved separately where the page allows
Skill tags (if shown)	Weak labels and vocabulary seeding	Kept apart from the gold annotations
Posting date / expiry	Period assignment	Missing values kept as missing
Location	Regional comparison	Mapped to broad regions
Company name	Cross-board deduplication	Used only for matching, then dropped from the analysis dataset
Salary (optional)	Descriptive only	Only if explicit
Source board	Platform-effect checks	Stored per posting

Quality and bias controls.

- Remove exact duplicates and flag cross-board reposts (title, company, location, and text similarity).
- Report the number of postings per board, role, city, and month; analyze results per board to show platform effects.
- Acknowledge that listing order, active-only visibility, and each board's employer mix bias the sample; do not describe the corpus as the whole market.
- Treat the manual corpus as the evaluation and annotation pool, with the historical dataset used for scale where it is available.

4.3. Project fields

Table 3: Fields used in the analysis dataset

Field	Purpose	Handling
Posting ID	Deduplication and joins	Replaced by a generated project ID
Job title	Assign IT/BA family	Normalized separately from description
Description / requirements	Skill extraction	Processed locally; contact details removed
Posting date	Trend summaries	ISO date, derived year/quarter; missingness kept
Location	Regional comparison	Mapped to broad regions; unknown kept separate
Language	Coverage analysis	Vietnamese / English / mixed label
Salary (optional)	Descriptive only	Used only if period and currency are explicit

4.4. Data preparation

1. Build a raw-data inventory and profile missingness, text length, and encoding issues by year and source.
2. Standardize core fields; remove exact duplicates and flag likely reposts using title, company, location, date proximity, and text similarity.
3. Normalize dates and locations; detect language (Vietnamese, English, mixed) and audit a sample manually.
4. Mask e-mail addresses, phone numbers, and personal names that are not needed for analysis.
5. Record row counts after every filter in a cleaning audit so the analysis can be reproduced.

5. Methodology

5.1. Pipeline overview

Figure 1 summarizes the proposed pipeline. Occupation classification (what kind of job a posting describes) and skill-span extraction (which skills a posting mentions) are kept as separate tasks so that each can be evaluated independently.

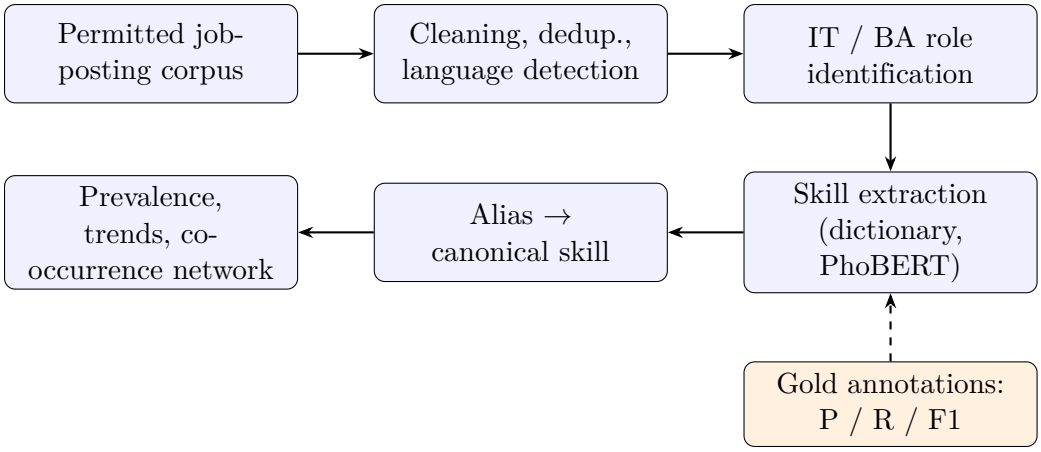


Figure 1: Proposed skill-mining pipeline

5.2. IT and BA job-family identification

A title taxonomy with aliases will be drafted, including rules for ambiguous roles (e.g. *Data Analyst* is not automatically a BA role). A stratified sample of titles will be labeled as IT, BA, both, other, or uncertain, with a second reviewer labeling a subset to measure agreement. The baseline is a set of transparent keyword rules; an optional classifier uses word/character TF–IDF with logistic regression or a linear SVM, or a fine-tuned PhoBERT [4] model (we may use another model or customize another baseline later) if enough labeled data is available. Results are reported as macro-F1, per-class precision/recall/F1, and a confusion matrix.

5.3. Skill vocabulary and annotation

A versioned bilingual vocabulary will record, for each entry, a canonical label, aliases, language, category (programming languages, frameworks, cloud/DevOps, data/AI, testing/security, databases, BA methods, requirements/process, tools, communication/domain skills), source, and ambiguity notes. ESCO [6] and O*NET [7] are used as supplementary references only, because neither is a Vietnam-specific taxonomy and ESCO has no Vietnamese language pack. A written annotation guide will define span boundaries, lists, acronyms, negation, and the distinction between required and preferred skills. A stratified sample (by role, period, and language) will be annotated, with a double-labeled subset used to report inter-annotator agreement.

5.4. Skill extraction models

Table 4: Extraction approaches compared

Approach	Role	Notes
Dictionary / phrase matching	Required baseline	Boundary-aware, interpretable, easy to audit; misses unseen forms
PhoBERT token classification (BIO)	Main neural model	Context-sensitive Vietnamese representation [4]; predicts exact spans
XLM-R token classification	Optional	For substantially mixed Vietnamese–English text [5]
CRF / BiLSTM-CRF	Optional	Traditional sequence-labeling comparison
Embedding-based linking	Optional	Suggests canonical labels for unseen phrases; always manually validated

Data are split by posting, keeping likely duplicates in the same split; if the sample allows, the latest year is reserved as a temporal test set. The untouched test set is used only for final reporting.

5.5. Evaluation of extraction

Let TP, FP, and FN denote true positives, false positives, and false negatives with respect to the adjudicated gold spans. We report

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}, \tag{1}$$

for exact-span and relaxed-overlap matching, together with canonical-skill micro and macro F1, per-category and per-language results, and support counts.

5.6. Prevalence and co-occurrence analysis

Let D_g be the eligible postings in group g (role, location, or period) and $S(d)$ the set of canonical skills in posting d. The main demand measure is posting prevalence:

$$\mathrm{prev}_g(s) = \frac{|\{\, d \in D_g : s \in S(d) \,\}|}{|D_g|}. \tag{2}$$

For skills a and b with posting sets D_a and D_b among N postings, co-occurrence strength is measured with

$$J(a,b) = \frac{|D_a \cap D_b|}{|D_a \cup D_b|}, \qquad \mathrm{lift}(a,b) = \frac{N\,|D_a \cap D_b|}{|D_a|\,|D_b|}. \tag{3}$$

Separate weighted, undirected networks will be built for IT and BA, with nodes as canonical skills and edges above a documented minimum support. Community detection and centrality are used as exploratory summaries, and clusters are interpreted through their member skills. Time series are aggregated by year or quarter and always shown with their denominators.

5.7. Baselines and robustness checks

- Dictionary-only extraction versus the final model.
- Exact matching without alias normalization versus normalized output.
- Alternative duplicate filters, minimum-text rules, and edge-support thresholds.
- Results by language subset and by source composition per period.

6. Success Criteria

The project will be considered successful if it:

1. produces a traceable, documented mapping from surface forms to canonical skills;
2. reports extraction precision, recall, and F1 on a manually reviewed sample;
3. provides interpretable comparisons of skill prevalence and co-occurrence for IT and BA postings;
4. documents sampling, data limitations, and processing decisions well enough for another student to reproduce the analysis.

No minimum accuracy threshold is assumed in advance; weak results will be reported and the analysis narrowed to reliably extracted skill groups.

7. Ethics and Responsible Interpretation

- Follow dataset licenses and source terms; document the basis for access and use.
- Publish aggregated results only; do not redistribute raw posting text unless explicitly permitted.
- Remove personal contact details; collect no data about applicants or employees.
- Describe findings as patterns in the sampled postings, and report coverage, language, platform, and deduplication biases.
- Do not interpret an increase in mentions as proof of a causal change in employer needs.

8. Tools and Implementation

Python with `pandas` (data preparation), regular expressions or a phrase matcher (dictionary baseline), Hugging Face `transformers` (PhoBERT/XLM-R), `scikit-learn` (classifiers and metrics), `networkx` (co-occurrence graphs), and `matplotlib`/ `seaborn` (visu-

alization). The skill vocabulary is stored as a versioned CSV/JSON file; configuration, random seeds, and parameters are kept under version control (Git) alongside the code.

9. Work Plan

Table 5: Nine-week timeline

Week	Activities	Milestone
1	Define scope, role families, time window; search candidate sources	Approved scope memo
2	License and provenance checks; inspect samples; select primary source	Source decision record
3	Data profiling, cleaning, deduplication, language detection	Clean dataset v1 and audit
4	Title taxonomy and labeling; pilot-read postings for skills	Role guide, pilot notes
5	Role rules/classifier; bilingual vocabulary; annotation of gold sample	Vocabulary v1, gold set
6	Dictionary baseline; BIO data preparation; PhoBERT fine-tuning	Model predictions
7	Extraction evaluation and error analysis; prevalence, trend, location, and network analyses	Evaluation table, figures
8	Robustness checks; draft methods, results, and limitations	Findings memo, report draft
9	Report integration, reproducibility review, presentation	Final report and slides

If time is short, the optional models (XLM-R, CRF) and salary analysis are dropped before any reduction of the manual evaluation.

10. Division of Work

Table 6: Team roles and responsibilities

Member	**Main responsibility**	**Outputs**
Nguyễn Trần Minh Trí (202400116)	Data and provenance	Source and license register, clean dataset, data-quality report, cleaning code
Tô Duy An (202416653)	IT/BA job-family classification	Role guide, labeled titles, rule baseline and classifier with metrics
Kiều Việt Sơn (202416750)	Skills and NLP	Bilingual vocabulary, gold annotations, dictionary and PhoBERT extraction, evaluation
Nguyễn Ngọc Tuấn Anh (202416659)	Mining and communication	Prevalence, trend, and network analyses; robustness checks; report and presentation integration

All members take part in the scope freeze, double annotation, results review, and final presentation.

11. Risks and Mitigation

Table 7: Main risks

Risk	Effect	Mitigation
Unclear license or provenance	Dataset unusable	Verify before analysis; switch to a permitted source
Missing years in the source	No valid trend analysis	Shorten the window or reframe as a snapshot
Short, missing, or duplicated text	Biased prevalence	Profile coverage, deduplicate, minimum-text rules, report exclusions
Ambiguous aliases, mixed language	Lower precision/recall	Boundary-aware matching, context rules, per-language evaluation
Inconsistent role labels	Mixed comparisons	Few families; validate rules on a labeled sample
Too few records per group	Unstable estimates	Merge groups, raise minimum support, or omit the comparison
Site terms forbid collection or the site blocks requests	Source unusable	Check terms first; stop on blocks; switch board or dataset
Snapshot-only boards	No historical trend	Report as a snapshot; use the historical dataset for years where it is verified

12. Expected Deliverables

1. A data statement covering source, license, coverage, sampling, and limitations.
2. A versioned bilingual skill vocabulary with aliases and ambiguity notes.
3. A reproducible cleaning, role-identification, extraction, and normalization pipeline.
4. A manually annotated gold sample with reported precision, recall, and F1.
5. Tables and figures of skill prevalence by role, location, and period.
6. IT and BA skill co-occurrence networks with interpretation.
7. A final report, presentation, and README with reproduction steps.

13. Conclusion

This project turns unstructured Vietnamese job-advertisement text into an auditable dataset of normalized skills, measures how reliably those skills can be extracted, and applies Web Mining methods to describe skill demand and skill combinations in IT and BA roles. Its focused scope, manual quality check, and explicit account of data provenance make it feasible within one semester while still producing measurable results.

References

[1] *VietJobs: A Vietnamese Job Advertisement Dataset*. arXiv preprint arXiv:2603.05262. https://arxiv.org/abs/2603.05262

[2] KALD Labs. *VieJobBERT: VietJobs-37K dataset and benchmark code*. https://github.com/kaldlabs/VieJobBERT

[3] Techmap. *Vietnam job posting data*. https://jobdatafeeds.com/data/countries/vn

[4] D. Q. Nguyen and A. T. Nguyen. PhoBERT: Pre-trained language models for Vietnamese. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 1037–1042, 2020.

[5] A. Conneau, K. Khandelwal, N. Goyal, et al. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 8440–8451, 2020.

[6] European Commission. *ESCO: European Skills, Competences, Qualifications and Occupations*. https://esco.ec.europa.eu/en/use-esco

[7] National Center for O*NET Development. *O*NET license agreements*. https://www.onetcenter.org/license_agreements.html